



Experimental Comparison Tables

1. VRAM Consumption (MB)

Lower is better.

Experiment Setup	512 Tokens	1024 Tokens	2048 Tokens	4096 Tokens
Exp 1: Vanilla MHA (24 Layers)	2094.65	2097.65	2099.65	3949.71
Exp 2: GQA + RoPE (24 Layers)	1870.76	1870.75	1870.75	3216.97
Exp 3: GTA + ALiBi (24 Layers)	2016.89	2016.89	2016.89	3177.60
Exp 4: GQA + RoPE (28 Layers)	2121.60	2121.13	2121.60	3376.95
Exp 5: GQA + RoPE (30 Layers)	2258.16	2258.16	2258.25	3593.08
Exp 6: GTA + ALiBi (28 Layers)	2213.35	2213.35	2213.35	3292.91
Exp 7: GTA + ALiBi (30 Layers)	2337.79	2337.79	2337.79	3463.92

2. Time to First Token (TTFT) - Pre-fill Latency (Seconds)

Lower is better. Measures how fast the prompt context is ingested.

Experiment Setup	512 Tokens	1024 Tokens	2048 Tokens	4096 Tokens
Exp 1: Vanilla MHA (24 Layers)	0.2447	0.2699	0.3140	0.5396
Exp 2: GQA + RoPE (24 Layers)	0.2637	0.2474	0.3551	0.6820
Exp 3: GTA + ALiBi (24 Layers)	0.2535	0.2547	0.2918	0.5832
Exp 4: GQA + RoPE (28 Layers)	0.2598	0.2634	0.3933	0.7680
Exp 5: GQA + RoPE (30 Layers)	0.3221	0.2742	0.4905	0.7424
Exp 6: GTA + ALiBi (28 Layers)	0.2556	0.2610	0.3269	0.6402
Exp 7: GTA + ALiBi (30 Layers)	0.2592	0.2718	0.3445	0.7286

3. Time Per Output Token (TPOT) - Decoding Throughput (Seconds)

Lower is better. Measures the continuous token generation speed.

Experiment Setup	512 Tokens	1024 Tokens	2048 Tokens	4096 Tokens
Exp 1: Vanilla MHA (24	0.0478	0.0514	0.0489	0.0514

Layers)				
Exp 2: GQA + RoPE (24 Layers)	0.0655	0.0664	0.0638	0.0682
Exp 3: GTA + ALiBi (24 Layers)	0.0497	0.0480	0.0494	0.0495
Exp 4: GQA + RoPE (28 Layers)	0.0705	0.0751	0.0748	0.0743
Exp 5: GQA + RoPE (30 Layers)	0.0805	0.0787	0.0830	0.0764
Exp 6: GTA + ALiBi (28 Layers)	0.0562	0.0564	0.0562	0.0586
Exp 7: GTA + ALiBi (30 Layers)	0.0571	0.0601	0.0603	0.0617

Key Extracted Insights

1. Static VRAM Plateaus vs. The 4K Context Cliff

- **The Context Floor:** For context sizes between 512 and 2048 tokens, VRAM remains practically identical for almost every single model variant. This reveals that the system is likely pre-allocating a static memory workspace, or the parameter weights dominate the base memory footprint at lower context windows.
- **The 4K Surge:** At 4096 tokens, every architecture hits a sharp threshold where VRAM consumption surges dramatically (jumping by roughly 1.1 GB to 1.8 GB). This indicates the transition point where KV-cache scaling overtakes static weight allocations.

2. Parameter Blueprint Nuance (MHA vs. RoPE/ALiBi)

- Looking closely at the reported parameter metrics, **Exp 1 (Vanilla MHA)** is the only architecture where total parameter size increases dynamically alongside context length (climbing from 354.3M to 357.9M).
- Conversely, every variant utilizing **GQA + RoPE** or **GTA + ALiBi** maintains a perfectly fixed parameter count regardless of context scaling. This highlights a structural difference: Vanilla MHA is likely instantiating explicit, sequence-length-dependent absolute position embeddings or static mask matrices, whereas RoPE and ALiBi calculate relative positional variations on the fly.

3. Decoding Throughput (TPOT): GTA Outpaces GQA

- **Decoding Speed is Prompt-Agnostic:** Across all 28 runs, the Time Per Output Token remains remarkably flat as prompt length increases. This proves the core optimization value of the KV-cache during generation.
- **GTA Throughput Efficiency:** Grounded Transformer Attention (GTA) combined with ALiBi is exceptionally fast during decoding. At 24 layers, GTA completely matches Vanilla MHA (~0.049s). Crucially, even when scaled up to a deep 30-layer configuration (**Exp 7**), GTA maintains a token decoding latency of **~0.060s**, making it faster than a significantly shallower 24-layer GQA configuration (**Exp 2** at ~0.065s).

4. Latency-Memory Trade-offs

- **The Pre-fill King:** Vanilla MHA maintains the fastest processing pipeline for context ingestion (lowest TTFT at 4096 tokens with 0.5396s).
- **The Efficiency Paradox:** Grouped Query Attention (GQA) dramatically optimizes long-context memory (a 30-layer GQA network consumes less VRAM at 4K context than a 24-layer Vanilla MHA network). However, this memory efficiency carries a performance tax: it records the slowest overall decoding throughput (TPOT) among equivalent layer sizes.